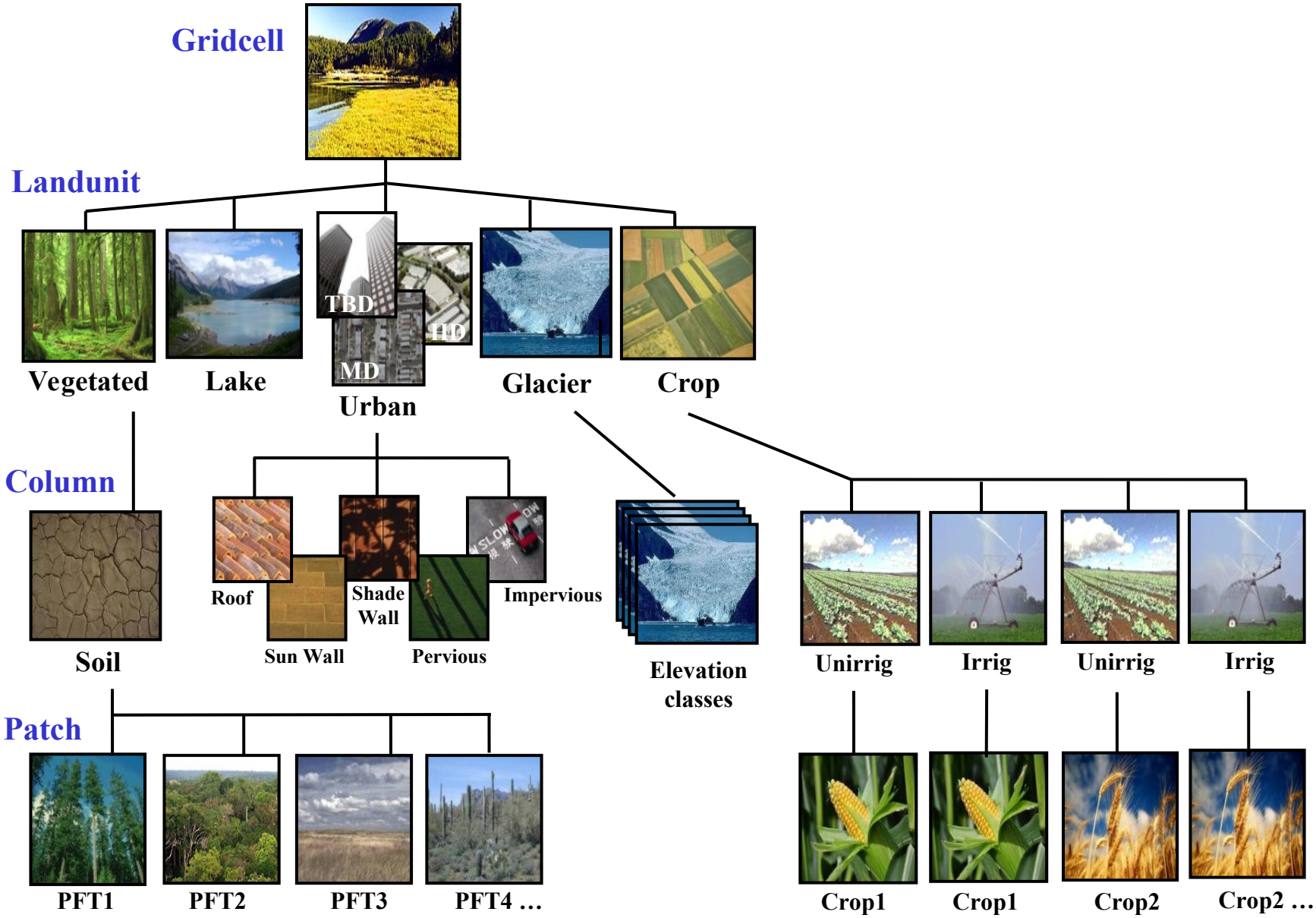
Hillslope Postprocessing

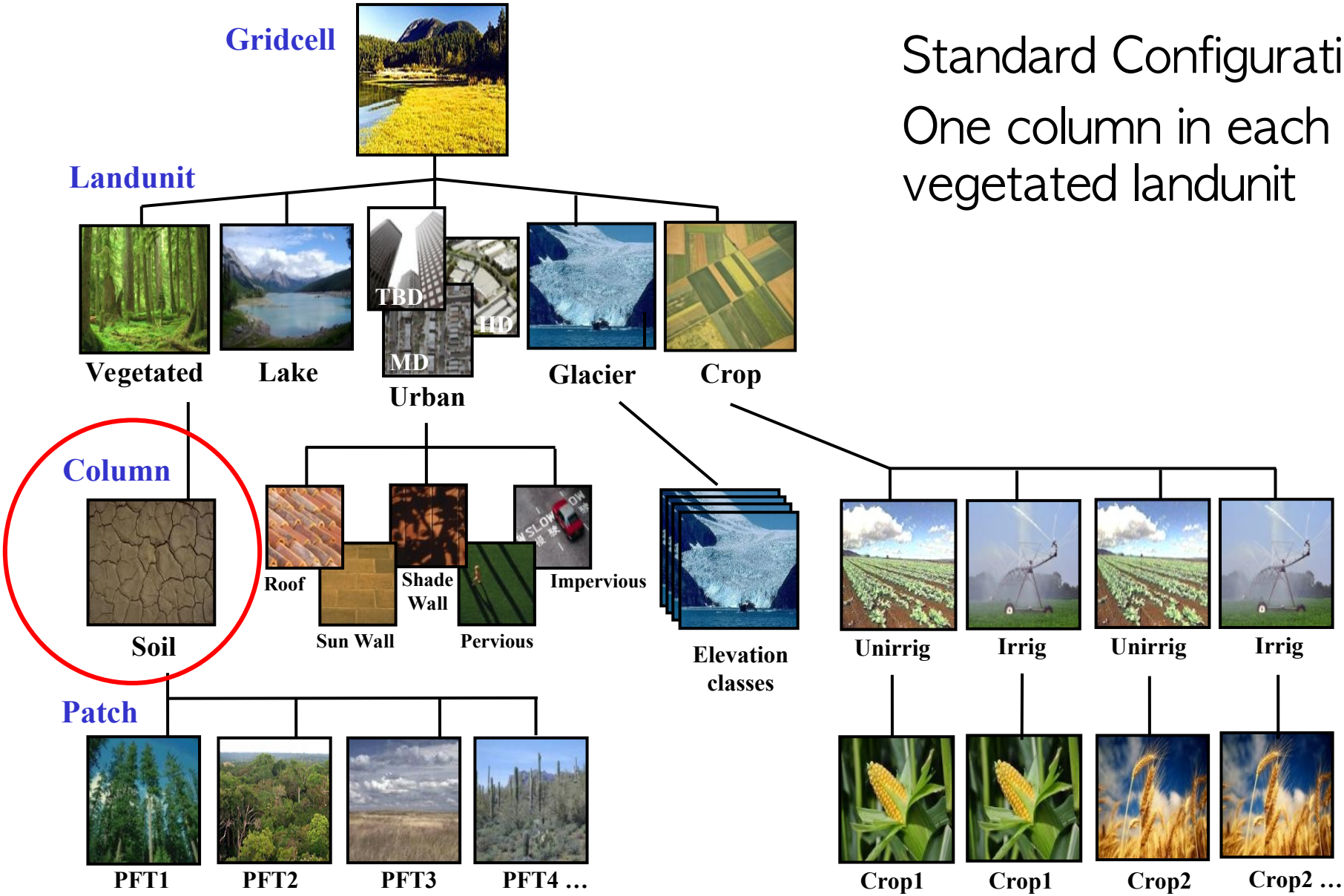
Individual hillslope column states and fluxes are saved to vector history files

To visualize column-level data (i.e. to create maps), it must be converted from vector (1d) to gridded (2d) format

CTSM Subgrid Tiling Structure



CTSM Subgrid Tiling Structure



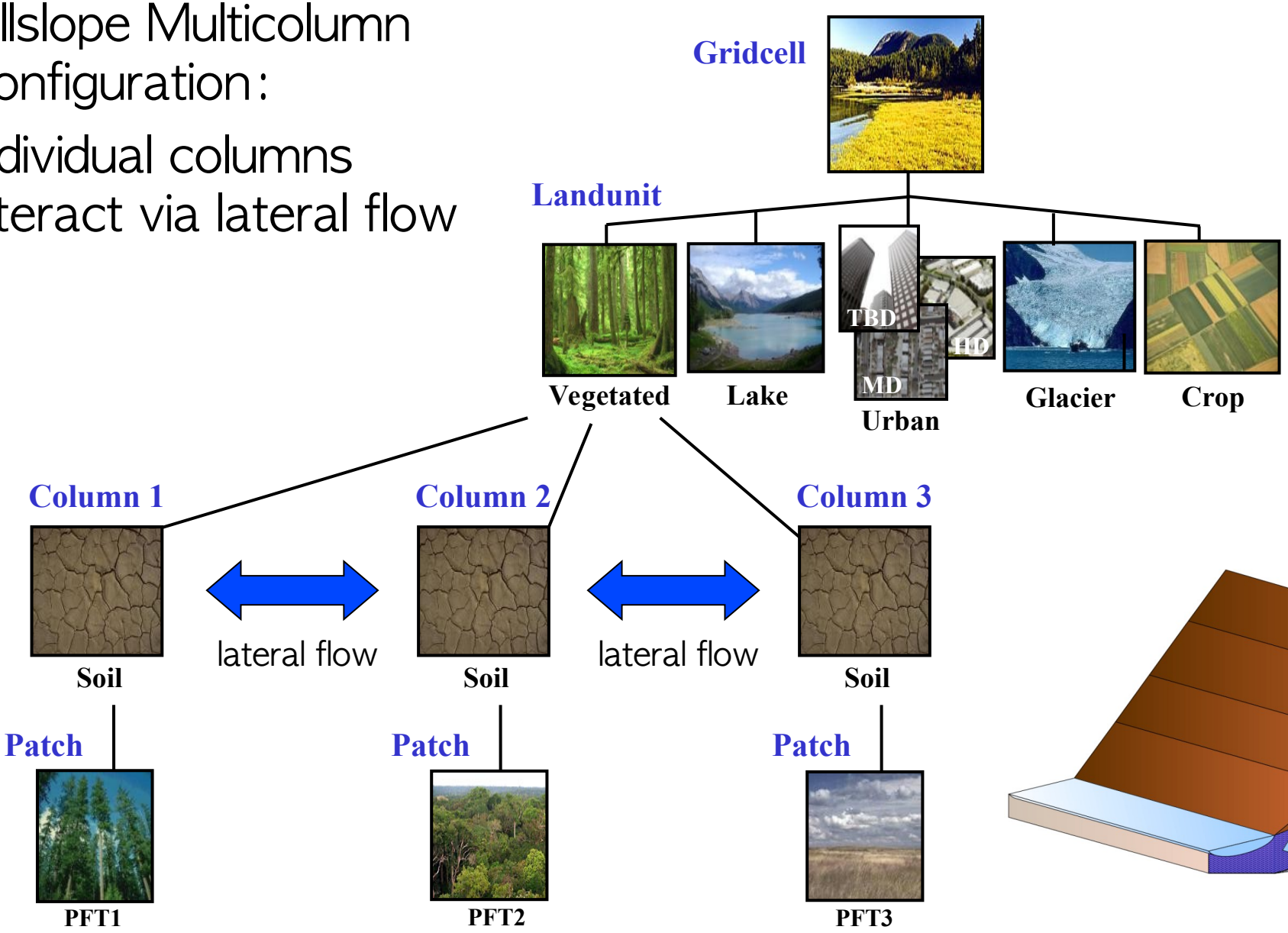
Standard Configuration:

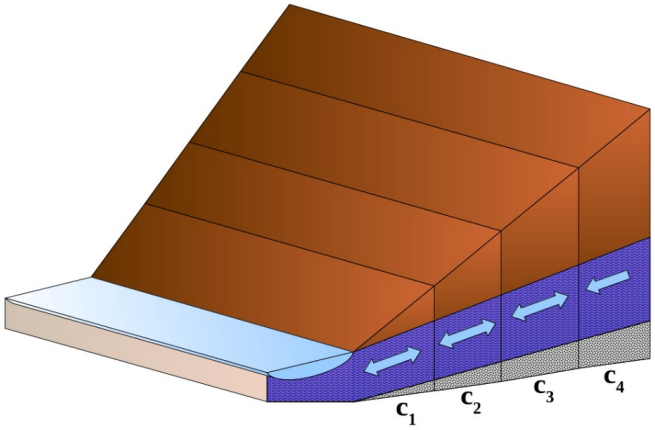
One column in each vegetated landunit

CTSM Subgrid Tiling Structure

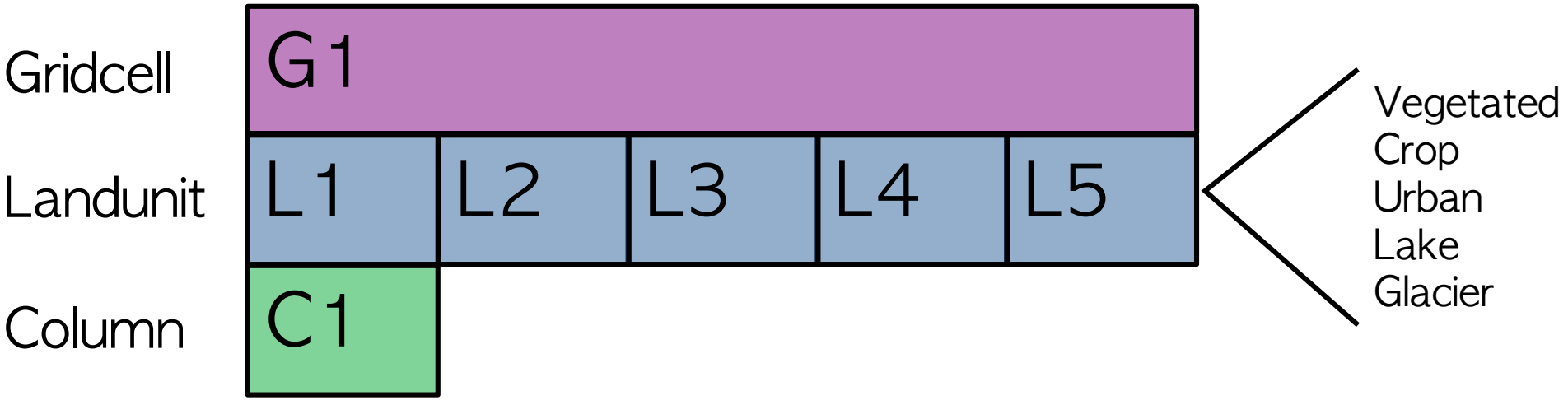
Hillslope Multicolumn Configuration:

Individual columns interact via lateral flow



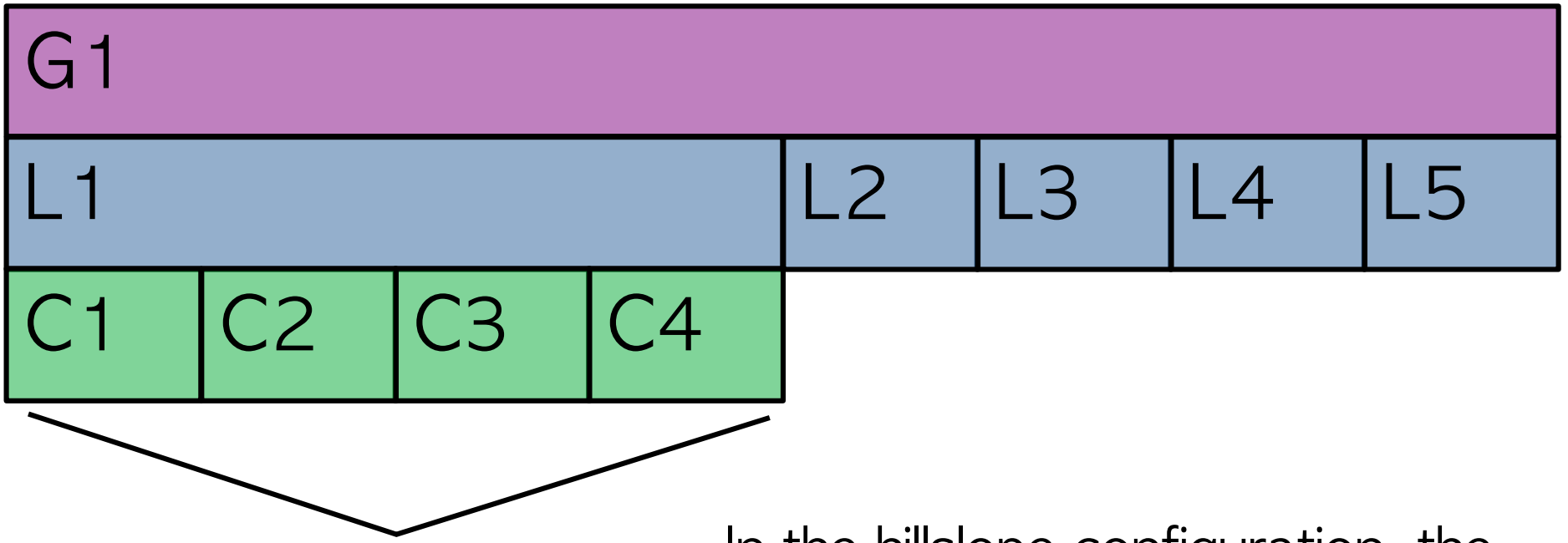


CTSM Vector Output



In the standard configuration, there is a 1-to-1 mapping from column to gridcell, so visualizing column-level variables is straightforward

CTSM Vector Output



In the hillslope configuration, the mapping from column to gridcell is degenerate, so visualizing column-level variables is less straightforward

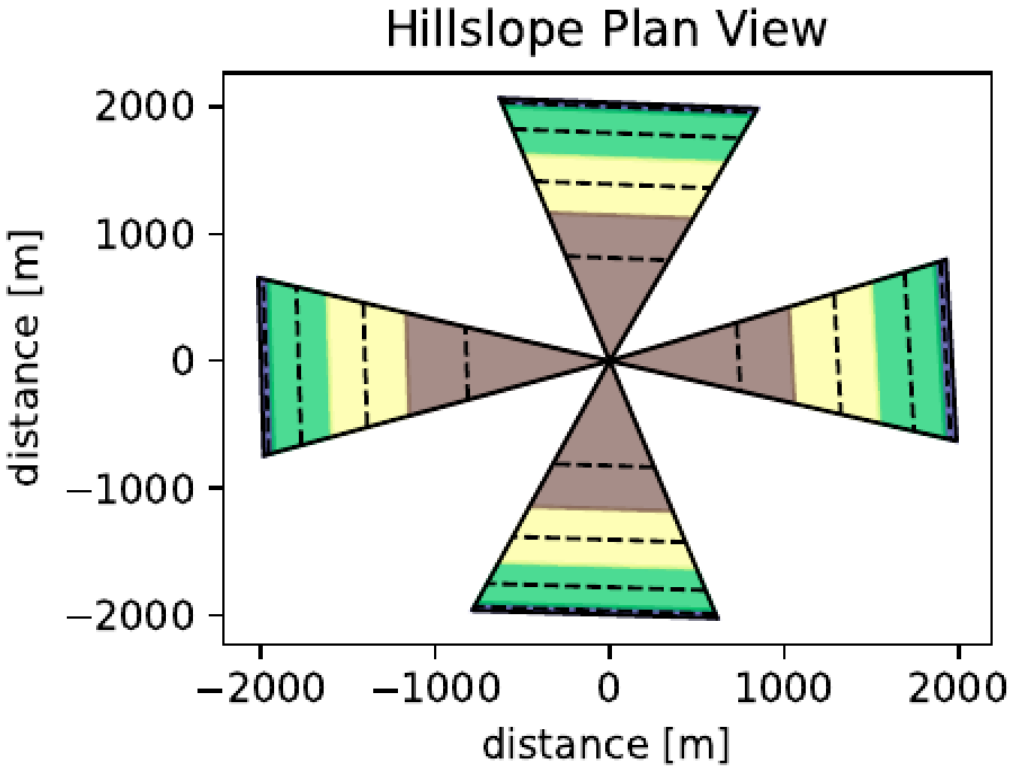
CTSM Vector Output

Vector-to-gridded workflow

Single column: north, lowland

Single hillslope (single aspect):
Requires area-weighting

All hillslope columns:
(exclude other landunits)

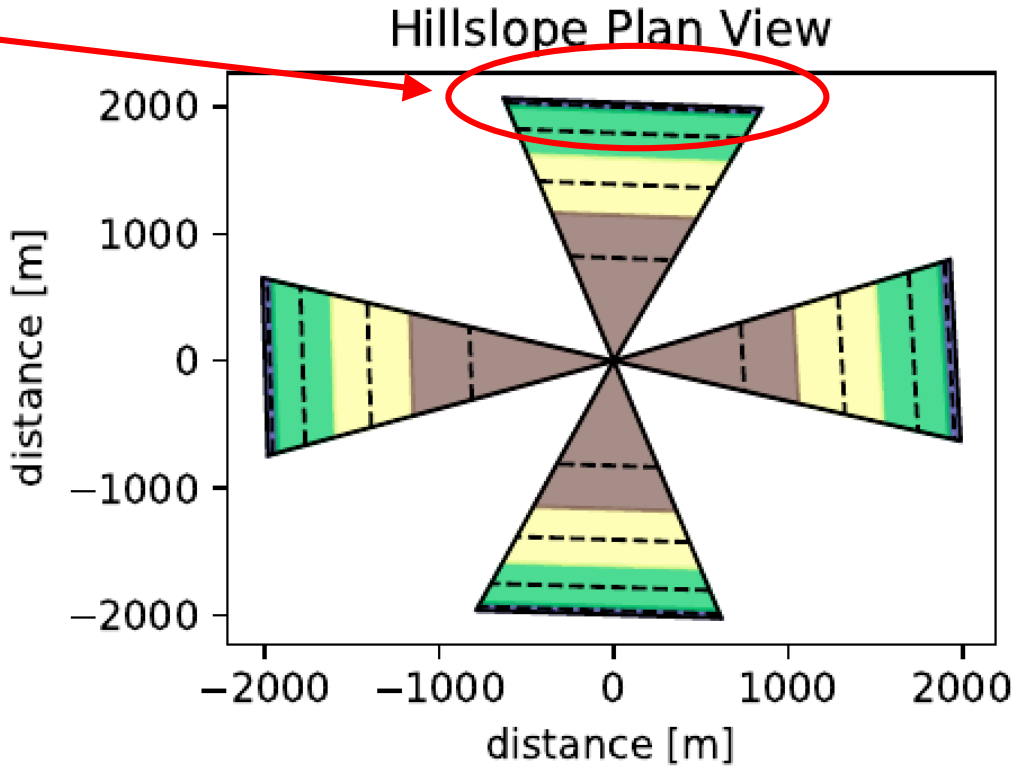


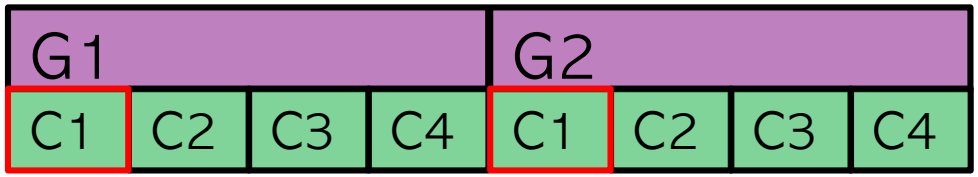
CTSM Vector Output

Vector-to-gridded workflow

Single column: north, lowland

Single hillslope (single aspect): Requires area-weighting





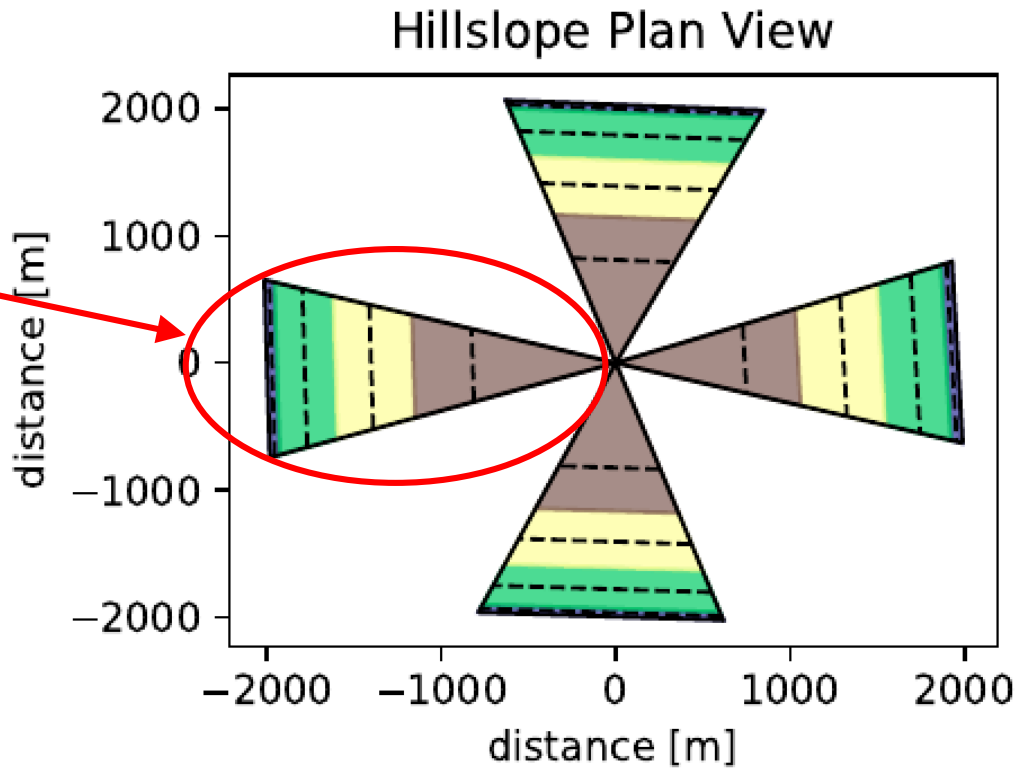
All columns having same aspect and position

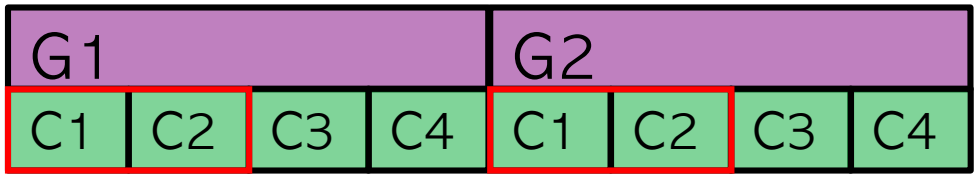
CTSM Vector Output

Vector-to-gridded workflow

Single column: north, lowland

Single hillslope (single aspect):
Requires area-weighting





Weighted sum of columns in gridcell 1

Weighted sum of columns in gridcell 2

Hillslope Postprocessing

1) convert all vector history files to gridded files for each subgrid unit (e.g. column or patch/pft)

2) convert vector to gridded as needed without saving new files

Hillslope Postprocessing

1) convert all vector history files to gridded files for each subgrid unit (e.g. column or patch/pft)

Pro: "one-and-done", all data available for use in standard visualization and analysis workflows

Con: large number of additional files with redundant information

Hillslope Postprocessing

2) convert vector to gridded as needed without saving new files

Pro: no storage overhead, less time required

Often useful when looking at a single gridcell

Con: may slow down workflow